TOPIC 4 · NAVIER-STOKES EQUATIONS

Solved Problems: Official Collection NS1 to NS4

Worked solutions to the Navier-Stokes collection of the Department of Thermal and Fluid Engineering, Universidad Carlos III de Madrid. Every problem restates the official statement word for word, then follows four phases: (1) hypotheses, data, unknowns and degrees of freedom; (2) coordinate system, form of the fields and boundary conditions; (3) the derivation, with continuity and each Navier-Stokes component simplified term by term, the integrations written out and every constant fixed with its boundary condition; (4) interpretation, limiting cases and the dimensional check. Where a computed result differs from the handwritten official solution, the discrepancy is flagged in the problem; anything beyond the course material is marked as an extension. References to Notes.pdf are to Chapter 4 (The Navier-Stokes equations) and Chapter 3 (Conservation laws).

PROBLEM NS1

Porous Wall with Suction

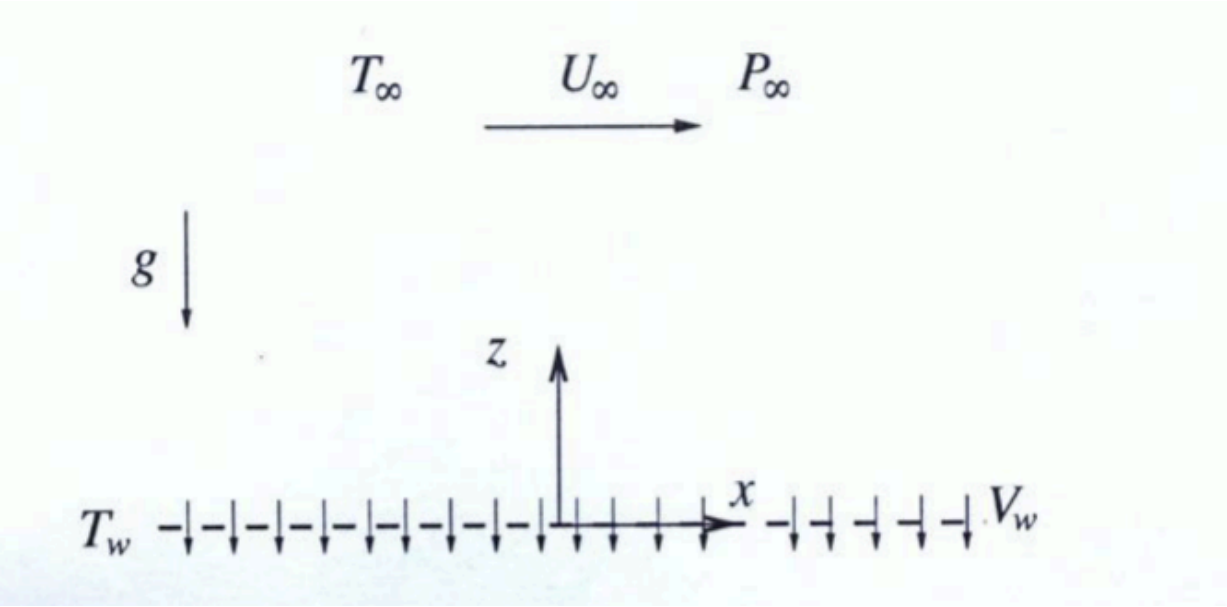
Original statement · NS1.pdf

Unidirectional flow | Porous wall | Viscous dissipation | Streamline

OFFICIAL PROBLEM STATEMENT

Consider the steady planar motion of a perfect liquid of density ρ, viscosity μ, thermal conductivity k and specific heat c in the vicinity of a horizontal porous wall. Far from the wall, the horizontal velocity component is $v_x = U_\infty$, the pressure is given by $p + \rho g z = P_\infty$ and the temperature is $T = T_\infty$, different from the value T_w at the wall. The liquid is sucked through the pores of the wall, resulting in a transverse velocity $v_z = -V_w$ at the wall surface. In the analysis, assume that the velocity and temperature fields are only a function of the distance to the wall z.

1. Determine the velocity components $v_x(z)$ and $v_z(z)$ and well as the pressure field $p(x, z)$.
2. Obtain the temperature field $T(z)$.
3. Calculate the streamline that crosses $x = x_o$ and $z = 0$.
4. Compute the viscous stress on the wall.
5. Determine the heat flux from the fluid to the wall.



Statement figure (crop of NS1.pdf, p. 1): horizontal porous wall at temperature T_w with suction velocity V_w, free stream U_∞, T_∞, P_∞, gravity g and axes x, z

`Source: NS1.pdf (page 1). Notes.pdf, Chapter 4: continuity Eq. 4.7, incompressible momentum Eq. 4.12, perfect-liquid energy Eq. 4.42, no-slip and far-field conditions (Section on initial and boundary conditions).`

Worked solution (4-phase method)

`PHASE 1`

Hypotheses, Data and Unknowns

- Steady planar flow of an incompressible liquid with constant ρ, μ, k, c (perfect liquid, Notes.pdf, Eq. 4.38): the equations to use are (4.40), (4.41) and (4.42).
- Gravity acts along $-\bar{e}_z$ (the far-field pressure is $p + \rho g z = P_\infty$, so $\bar{f}_m = -g\,\bar{e}_z$).
- Given: the fields depend only on the distance to the wall, $\partial/\partial x = \partial/\partial y = \partial/\partial t = 0$ for every variable, except that the pressure is allowed to depend on x (it is determined below).
- No chemical reaction and no radiation: $Q_c = Q_r = 0$.
- The wall is at rest and porous: the tangential velocity is zero (no-slip) and the normal velocity is the suction velocity.
- Data: $[\rho,\ \mu,\ k,\ c,\ U_\infty,\ P_\infty,\ T_\infty,\ T_w,\ V_w,\ g]$ with $V_w > 0$. Unknowns: $v_x(z), v_z(z)$, $p(x,z), T(z)$, the streamline, the wall stress and the wall heat flux.

- Degrees of freedom: two velocity components, the pressure and the temperature, each a function of one coordinate (plus x for the pressure): four unknown fields against continuity, two momentum components and the energy equation.

Symbol	Meaning	SI unit
$\rho, \mu, \nu = \mu/\rho$	density, dynamic and kinematic viscosity	kg/m^3, Pa s, m^2/s
$k, c, \alpha = k/(\rho c)$	thermal conductivity, specific heat, thermal diffusivity	W/(m K), J/(kg K), m^2/s
$\mathrm{Pr} = \nu/\alpha = \mu c/k$	Prandtl number	1
U_∞, V_w	free-stream velocity, suction velocity	m/s
P_∞, T_∞, T_w	far-field reduced pressure, far-field and wall temperatures	Pa, K, K

PHASE 2

Frames, Field Form and Boundary Conditions

- Cartesian frame (x, y, z) with x along the wall (in the direction of U_∞), z normal to it and pointing into the liquid, y out of the plane. The basis $\{\bar{e}_x, \bar{e}_y, \bar{e}_z\}$ is orthonormal and right-handed, so no change of basis is needed (det $R = 1$, $RR^T = I$ with $R = I$).
- Field form: $\bar{v} = v_x(z)\,\bar{e}_x + v_z(z)\,\bar{e}_z, T = T(z), p = p(x, z)$.
- The wall is at $z = 0$ and the liquid occupies $z > 0$.

Location	Condition	Origin
$z = 0$	$v_x = 0$	no-slip on the wall at rest
$z = 0$	$v_z = -V_w$	suction through the pores (statement)
$z = 0$	$T = T_w$	thermal equilibrium at the wall
$z \to \infty$	$v_x \to U_\infty$	free stream (statement)
$z \to \infty$	$p + \rho g z \to P_\infty$	free stream (statement)
$z \to \infty$	$T \to T_\infty$	free stream (statement)

PHASE 3

Step-by-Step Derivation

3.1 Continuity and the normal velocity

$$\nabla \cdot \bar{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_z}{\partial z} = \underbrace{\cancel{\frac{\partial v_x}{\partial x}}}_{v_x = v_x(z)} + \frac{dv_z}{dz} = 0 \implies \frac{dv_z}{dz} = 0 \implies v_z = \text{cons} \quad (1)$$

The constant is fixed at the wall: $v_z(0) = -V_w$, therefore

$$\boxed{v_z(z) = -V_w} \quad (2)$$

3.2 Momentum equations

For $\bar{f}_m = -g\,\bar{e}_z$ the Cartesian components of (4.41) with constant μ are (Notes.pdf, Section on the Navier-Stokes equations in Cartesian coordinates). Every term that contains $\partial/\partial t$, $\partial/\partial x$ or $\partial/\partial y$ of a velocity component is zero.

z-component:

$$\rho\Big(\underbrace{\cancel{\frac{\partial v_z}{\partial t}}}_{\text{steady}} + \underbrace{\cancel{v_x \frac{\partial v_z}{\partial x}}}_{v_z = v_z(z)} + v_z \underbrace{\frac{dv_z}{dz}}_{=0 \text{ by } (1)}\Big) = -\frac{\partial p}{\partial z} + \mu \underbrace{\frac{d^2 v_z}{dz^2}}_{=0} - \rho g \implies \frac{\partial p}{\partial z} = -\rho g \quad (3)$$

Integrating (3) with respect to z at fixed x (the "constant" may depend on x):

$$p + \rho g z = f(x) \quad (4)$$

The far-field condition $p + \rho g z \to P_\infty$ holds for every x, with P_∞ a constant, so $f(x) = P_\infty$:

$$\boxed{p(x,z) = P_\infty - \rho g z}, \qquad \frac{\partial p}{\partial x} = 0 \tag{5}$$

x-component:

$$\rho\Big(\underbrace{\cancel{\frac{\partial v_x}{\partial t}}}_{\text{steady}} + \underbrace{\cancel{v_x\frac{\partial v_x}{\partial x}}}_{\partial_x = 0} + v_z\frac{dv_x}{dz}\Big) = -\underbrace{\frac{\partial p}{\partial x}}_{=0 \text{ by } (5)} + \mu\frac{d^2 v_x}{dz^2} \implies -\rho V_w \frac{dv_x}{dz} = \mu\frac{d^2 v_x}{dz^2} \tag{6}$$

Dividing by μ and using $\nu = \mu/\rho$:

$$\frac{d^2 v_x}{dz^2} + \frac{V_w}{\nu}\frac{dv_x}{dz} = 0 \tag{7}$$

This is a linear ODE with constant coefficients. Trying $v_x = e^{\lambda z}$ gives $\lambda^2 + (V_w/\nu)\lambda = 0$, so $\lambda = 0$ or $\lambda = -V_w/\nu$:

$$v_x = C_1 + C_2\, e^{-V_w z/\nu} \tag{8}$$

Boundary conditions: $v_x(\infty) = U_\infty$ gives $C_1 = U_\infty$ (the exponential vanishes because $V_w > 0$); $v_x(0) = 0$ gives $C_1 + C_2 = 0$, so $C_2 = -U_\infty$:

$$\boxed{v_x(z) = U_\infty\left(1 - e^{-V_w z/\nu}\right)} \tag{9}$$

3.3 Energy equation and the temperature field

With $e = e_o + cT$, $\bar{v} = v_x \bar{e}_x + v_z \bar{e}_z$ and $\partial/\partial x = \partial/\partial t = 0$, (4.42) reads

$$\rho c\, v_z \frac{dT}{dz} = \phi_v + k \frac{d^2T}{dz^2} \tag{10}$$

The viscous dissipation of this flow, from the Cartesian expression of ϕ_v (Notes.pdf, Section on the Navier-Stokes equations in Cartesian coordinates), has a single non-zero shear rate $\partial v_x/\partial z$:

$$\phi_v = \mu \left(\frac{dv_x}{dz} \right)^2, \qquad \frac{dv_x}{dz} = \frac{U_\infty V_w}{\nu}\, e^{-V_w z/\nu} \implies \phi_v = \frac{\mu U_\infty^2 V_w^2}{\nu^2}\, e^{-2V_w z/\nu} \tag{11}$$

Substituting $v_z = -V_w$ and (11) in (10) and dividing by k, with $\alpha = k/(\rho c)$:

$$\frac{d^2T}{dz^2} + \frac{V_w}{\alpha}\frac{dT}{dz} = -\frac{\mu U_\infty^2 V_w^2}{k\,\nu^2}\, e^{-2V_w z/\nu} \tag{12}$$

The solution is the sum of the homogeneous solution $B_1 + B_2 e^{-V_w z/\alpha}$ and a particular solution $A\, e^{-2V_w z/\nu}$. Substituting the particular form in (12), with $T_p' = -(2V_w/\nu) A\, e^{-2V_w z/\nu}$ and $T_p'' = (4V_w^2/\nu^2) A\, e^{-2V_w z/\nu}$:

$$A\left(\frac{4V_w^2}{\nu^2} - \frac{2V_w^2}{\alpha \nu} \right) = -\frac{\mu U_\infty^2 V_w^2}{k \nu^2} \implies A\,(4 - 2\,\mathrm{Pr}) = -\frac{\mu U_\infty^2}{k} = -\frac{\mathrm{Pr}\, U_\infty^2}{c} \tag{13}$$

where $\mathrm{Pr} = \nu/\alpha = \mu c/k$ was used. Hence (for $\mathrm{Pr} \neq 2$)

$$A = \frac{\mathrm{Pr}}{2(\mathrm{Pr} - 2)} \frac{U_\infty^2}{c} \tag{14}$$

so that $T = B_1 + B_2 e^{-V_w z/\alpha} + A\, e^{-2V_w z/\nu}$. The condition $T(\infty) = T_\infty$ gives $B_1 = T_\infty$ and $T(0) = T_w$ gives $B_2 = T_w - T_\infty - A$:

$$\boxed{T(z) = T_\infty + \left(T_w - T_\infty - \frac{\mathrm{Pr}\, U_\infty^2}{2(\mathrm{Pr}-2)\,c}\right) e^{-V_w z/\alpha} + \frac{\mathrm{Pr}\, U_\infty^2}{2(\mathrm{Pr}-2)\,c}\, e^{-2V_w z/\nu}} \tag{15}$$

Note that $V_w z/\alpha = \mathrm{Pr}\, V_w z/\nu$. For $\mathrm{Pr} = 2$ the two exponents coincide and the particular solution takes the form $Az\, e^{-2V_w z/\nu}$ (a resonant case, not worked out in the sheet).

3.4 Streamline through $(x_o, 0)$

The streamlines are the integral curves of $\bar{v}$: $dx/v_x = dz/v_z$. With (2) and (9):

$$\frac{dx}{dz} = \frac{v_x}{v_z} = -\frac{U_\infty}{V_w}\left(1 - e^{-V_w z/\nu}\right) \tag{16}$$

Integrating from $z' = 0$ (where $x = x_o$) to $z' = z$. The antiderivative of $1 - e^{-az'}$ with $a = V_w/\nu$ is $z' + e^{-az'}/a$, and Barrow's rule gives $\left[z' + e^{-az'}/a\right]_0^z = z + (e^{-az} - 1)/a$:

$$\boxed{x - x_o = -\frac{U_\infty}{V_w}\left[z + \frac{\nu}{V_w}\left(e^{-V_w z/\nu} - 1\right)\right]} \tag{17}$$

3.5 Viscous stress on the wall

The wall is the surface $z = 0$. Take the unit normal pointing into the liquid, $\bar{n} = \bar{e}_z$: then $\bar{\bar{\tau}}' \cdot \bar{n}$ is the viscous force per unit area that the liquid exerts on the wall. The components of $\bar{\bar{\tau}}'$ with the Navier-Poisson law (Notes.pdf, Section on stress components, Cartesian coordinates) are

$$\bar{\bar{\tau}}' \cdot \bar{e}_z = \tau'_{xz}\,\bar{e}_x + \tau'_{zz}\,\bar{e}_z, \qquad \tau'_{xz} = \mu\left(\frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x}\right) = \mu\frac{dv_x}{dz}, \qquad \tau'_{zz} = 2\mu\frac{dv_z}{dz} + \left(\mu_B - \tfrac{2}{3}\mu\right)\nabla\cdot\bar{v} = (\tag{18}$$

At the wall, from (9), $dv_x/dz\big|_{z=0} = U_\infty V_w/\nu$, so

$$\boxed{\bar{\bar{\tau}}' \cdot \bar{n}\Big|_w = \mu \frac{U_\infty V_w}{\nu}\, \bar{e}_x = \rho V_w U_\infty\, \bar{e}_x} \tag{19}$$

The liquid drags the wall in the direction of the free stream.

3.6 Heat flux from the liquid to the wall

The heat flux from the liquid to the wall is $\bar{q} \cdot \bar{n}$ with $\bar{n} = -\bar{e}_z$ (outward from the liquid, into the wall), and $\bar{q} = -k\nabla T$ (Notes.pdf, Eq. 4.36):

$$\bar{q} \cdot \bar{n}\Big|_w = -k \frac{dT}{dz} \bar{e}_z \cdot (-\bar{e}_z)\Big|_{z=0} = k \frac{dT}{dz}\Big|_{z=0} \tag{20}$$

Differentiating (15) and evaluating at $z = 0$:

$$\frac{dT}{dz}\Big|_0 = -\frac{V_w}{\alpha} B_2 - \frac{2V_w}{\nu} A = -\frac{V_w}{\nu} \left(\mathrm{Pr}\, B_2 + 2A\right) \tag{21}$$

Using $k/\nu = \rho c/\mathrm{Pr}$ and $B_2 = T_w - T_\infty - A$:

$$k \frac{dT}{dz}\Big|_0 = -\rho c V_w \left(B_2 + \frac{2A}{\mathrm{Pr}}\right) = -\rho c V_w \left[T_w - T_\infty - A\,\frac{\mathrm{Pr} - 2}{\mathrm{Pr}}\right] = -\rho c V_w \left[T_w - T_\infty - \frac{U_\infty^2}{2c}\right] \tag{22}$$

$$\boxed{\bar{q} \cdot \bar{n}\Big|_w = \rho c V_w \left[T_\infty - T_w + \frac{U_\infty^2}{2c}\right]} \tag{23}$$

The result does not depend on the Prandtl number.

PHASE 4

Interpretation, Limits and Dimensional Check

- **Physical meaning.** Suction pins the shear layer to the wall: the velocity reaches 99% of U_∞ at $z \approx 4.6\,\nu/V_w$ and the thickness ν/V_w does not grow with x. The liquid is pulled towards the wall by the pores, and the viscous diffusion away from the wall balances this convection (equation (6)).
- **Limits.** For $V_w \to 0$ the layer thickens without bound, consistent with the absence of a steady unidirectional solution. For $z \to \infty$, $v_x \to U_\infty$ and $T \to T_\infty$. At $z = 0$, $v_x = 0$ and $T = T_w$.
- **Dimensional check.** $V_w z/\nu$ and $V_w z/\alpha$ are dimensionless. In (23), $\rho c V_w\,\Delta T$ has units $(\mathrm{kg/m^3})(\mathrm{J/(kg\,K)})(\mathrm{m/s})\,\mathrm{K} = \mathrm{W/m^2}$, and $U_\infty^2/(2c)$ has units $(\mathrm{m^2/s^2})/(\mathrm{J/(kg\,K)}) = \mathrm{K}$. In (19), $\rho V_w U_\infty$ has units $\mathrm{kg/(m^2\,s)} \cdot \mathrm{m/s} = \mathrm{N/m^2}$.
- **Energy check.** Equation (23) is the heat that must flow into the wall in steady state: per unit area the suction brings in from the free stream the energy $\rho V_w (c T_\infty + U_\infty^2/2)$ and the pores take out $\rho V_w\, c T_w$; the pressure work and the work of gravity cancel, and the shear does no work on the wall because it is at rest. The difference is $\rho c V_w [T_\infty - T_w + U_\infty^2/(2c)]$, the result (23).
- **Special case.** For $U_\infty \to 0$ there is no dissipation and (23) reduces to $\rho c V_w (T_\infty - T_w)$, the advection-conduction balance of pure suction.

PROBLEM NS2

Rotating Coaxial Cylinders

Original statement · NS2.pdf

Circular Couette flow | Torque and power | Viscous dissipation | Temperature profile

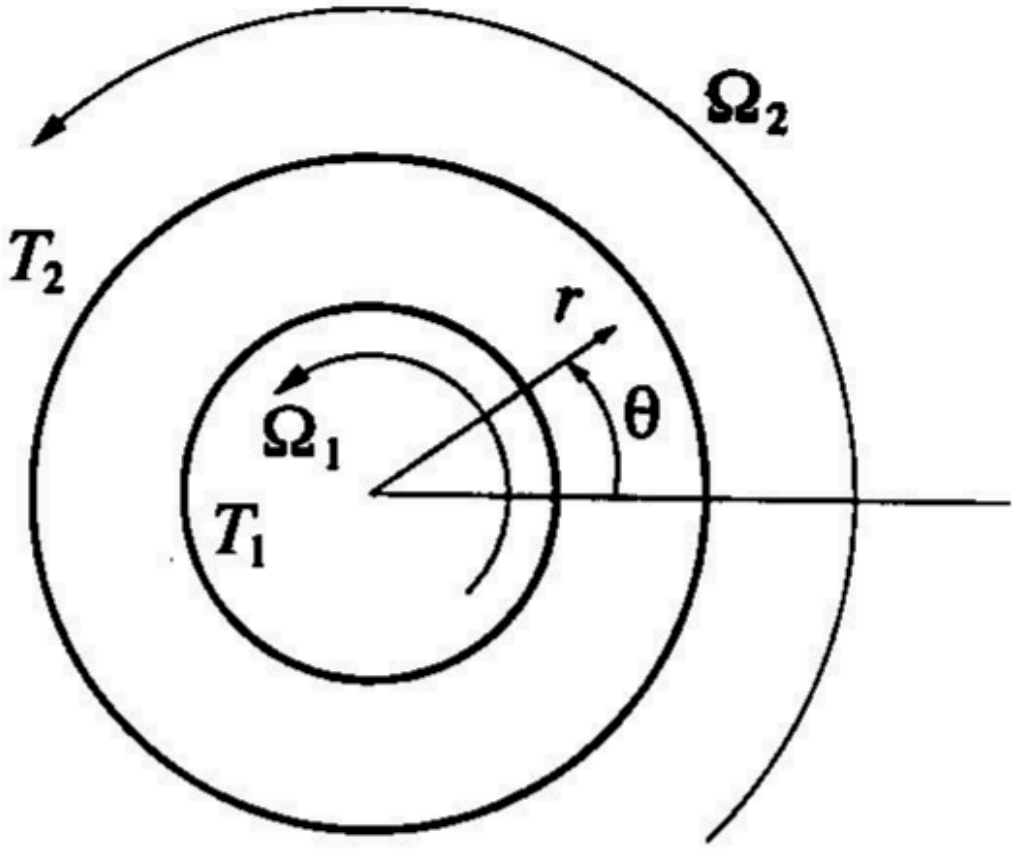
OFFICIAL PROBLEM STATEMENT

Consider the steady planar motion of a perfect liquid of density ρ, viscosity μ, thermal conductivity k and specific heat c in the gap formed between two coaxial cylinders of radii R_1 and R_2 that are rotating with angular velocities Ω_1 and Ω_2, respectively. Because of the symmetry of the problem, $v_z = 0$ and $\partial/\partial z = \partial/\partial\theta = 0$.

1. Obtain the velocity and pressure fields.

2. Calculate the torque exerted on each cylinder, as well as the power needed to keep them rotating.

3. If the temperatures on the cylinder surfaces are T_1 and T_2, determine the temperature distribution $T(r)$.



Statement figure (crop of NS2.pdf, p. 1): inner cylinder R_1 at T_1 rotating with Ω_1, outer cylinder R_2 at T_2 rotating with Ω_2, polar coordinates r, θ

Source: NS2.pdf (page 1). Notes.pdf, Chapter 4: continuity Eq. 4.40, perfect-liquid momentum Eq. 4.41 in cylindrical coordinates, energy Eq. 4.42, stress components in cylindrical coordinates.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady planar motion ($v_z = 0, \partial/\partial z = 0$) of a perfect liquid with constant ρ, μ, k, c: equations (4.40)-(4.42) in cylindrical coordinates.
- Axisymmetric: $\partial/\partial\theta = 0$ for every variable.

- No mass force enters the problem in the plane of motion (the statement lists none; gravity, if present, only shifts the pressure by a hydrostatic term and is absorbed in the reduced pressure of Notes.pdf, Eq. 4.13). No chemical reaction and no radiation: $Q_c = Q_r = 0$.
- The cylinder surfaces are impermeable and move with their own angular velocity (no-slip).
- Data: $[\rho,\ \mu,\ k,\ c,\ R_1,\ R_2,\ \Omega_1,\ \Omega_2,\ T_1,\ T_2]$ with $R_2 > R_1$. Unknowns: $v_r(r), v_\theta(r)$, $p(r)$, the torques, the powers and $T(r)$.
- Degrees of freedom: two velocity components, the pressure and the temperature, each a function of r alone.

Symbol	Meaning	SI unit
R_1, R_2	inner and outer radii	m
Ω_1, Ω_2	angular velocities (positive counter-clockwise, towards increasing θ)	rad/s
T_1, T_2	temperatures of the inner and outer surfaces	K
ρ, μ, k	density, viscosity, thermal conductivity	kg/m^3, Pa s, W/(m K)
C_1, C_2	integration constants of v_θ	1/s, m^2/s

PHASE 2

Frames, Field Form and Boundary Conditions

- Cylindrical frame (r, θ, z) centred on the common axis, with $\bar{e}_r, \bar{e}_\theta, \bar{e}_z$ orthonormal and right-handed ($\bar{e}_r \wedge \bar{e}_\theta = \bar{e}_z$); the frame is attached to the laboratory, which is inertial.
- Field form: $\bar{v} = v_r(r)\,\bar{e}_r + v_\theta(r)\,\bar{e}_\theta, p = p(r), T = T(r)$.

Location	Condition	Origin
$r = R_1$	$v_r = 0, v_\theta = \Omega_1 R_1$	impermeable inner cylinder, no-slip
$r = R_2$	$v_r = 0, v_\theta = \Omega_2 R_2$	impermeable outer cylinder, no-slip
$r = R_1$	$T = T_1$	thermal equilibrium at the wall
$r = R_2$	$T = T_2$	thermal equilibrium at the wall

PHASE 3

Step-by-Step Derivation

3.1 Continuity and the radial velocity

$$\frac{1}{r}\frac{d(r v_r)}{dr} + \underbrace{\frac{1}{r}\frac{\partial v_\theta}{\partial \theta}}_{\partial_\theta = 0} + \underbrace{\frac{\partial v_z}{\partial z}}_{v_z = 0} = 0 \implies r\, v_r = \text{const} \tag{1}$$

The constant is fixed at the inner wall, where $v_r(R_1) = 0$: $r\, v_r = 0$ for every r, therefore

$$\boxed{v_r(r) = 0} \tag{2}$$

3.2 Azimuthal momentum and the velocity profile

The θ component of (4.41) in cylindrical coordinates, with $v_r = 0, \partial/\partial t = \partial_\theta = \partial_z = 0$ and no mass force:

$$\rho\Big(\underbrace{\frac{\partial v_\theta}{\partial t}}_{\text{steady}} + \underbrace{v_r\frac{\partial v_\theta}{\partial r}}_{v_r=0} + \underbrace{\frac{v_\theta}{r}\frac{\partial v_\theta}{\partial \theta}}_{\partial_\theta=0} + \underbrace{\frac{v_r v_\theta}{r}}_{v_r=0} + \underbrace{v_z\frac{\partial v_\theta}{\partial z}}_{v_z=0}\Big) = -\underbrace{\frac{1}{r}\frac{\partial p}{\partial \theta}}_{\partial_\theta=0} + \mu\frac{d}{dr}\left[\frac{1}{r}\frac{d(r v_\theta)}{dr}\right] \tag{3}$$

Hence $\dfrac{d}{dr}\left[\dfrac{1}{r}\dfrac{d(r v_\theta)}{dr}\right] = 0$. Integrating once, $\dfrac{1}{r}\dfrac{d(r v_\theta)}{dr} = 2C_1$ (the factor 2 is a convenient choice of constant). Multiplying by r and integrating again, $r v_\theta = C_1 r^2 + C_2$:

$$v_\theta = C_1\, r + \frac{C_2}{r} \tag{4}$$

The boundary conditions $v_\theta(R_1) = \Omega_1 R_1$ and $v_\theta(R_2) = \Omega_2 R_2$, each multiplied by R_i, give a linear system:

$$C_1 R_1^2 + C_2 = \Omega_1 R_1^2, \qquad C_1 R_2^2 + C_2 = \Omega_2 R_2^2 \tag{5}$$

Subtracting the first equation from the second, $C_1 (R_2^2 - R_1^2) = \Omega_2 R_2^2 - \Omega_1 R_1^2$, and then $C_2 = \Omega_1 R_1^2 - C_1 R_1^2$:

$$C_1 = \frac{\Omega_2 R_2^2 - \Omega_1 R_1^2}{R_2^2 - R_1^2}, \qquad C_2 = \frac{(\Omega_1 - \Omega_2)\, R_1^2 R_2^2}{R_2^2 - R_1^2} \tag{6}$$

$$\boxed{v_\theta(r) = \frac{\Omega_2 R_2^2 - \Omega_1 R_1^2}{R_2^2 - R_1^2}\, r + \frac{(\Omega_1 - \Omega_2) R_1^2 R_2^2}{R_2^2 - R_1^2}\, \frac{1}{r}} \tag{7}$$

3.3 Radial momentum and the pressure field

The r component of (4.41) with $v_r = 0$ keeps only the centrifugal term and the pressure gradient (all other terms vanish for the same reasons as in (3)):

$$-\rho \frac{v_\theta^2}{r} = -\frac{dp}{dr} \quad \Longrightarrow \quad \frac{dp}{dr} = \rho \frac{v_\theta^2}{r} = \rho \left(C_1^2\, r + \frac{2 C_1 C_2}{r} + \frac{C_2^2}{r^3} \right) \tag{8}$$

Integrating from R_1 to r, the antiderivatives are $C_1^2 r^2/2$, $2C_1C_2 \ln r$ and $-C_2^2/(2r^2)$, and Barrow's rule gives

$$\boxed{p(r) - p(R_1) = \rho \left[\frac{C_1^2}{2} \left(r^2 - R_1^2 \right) + 2 C_1 C_2 \ln \frac{r}{R_1} - \frac{C_2^2}{2} \left(\frac{1}{r^2} - \frac{1}{R_1^2} \right) \right]} \tag{9}$$

The constant $p(R_1)$ is set by the reference pressure of the problem; the pressure grows outwards because the fluid is whirling and needs a centripetal force.

3.4 Viscous stress and torque on each cylinder

The only non-zero viscous shear stress is (Notes.pdf, stress components in cylindrical coordinates, with $v_r = 0$):

$$\tau'_{r\theta} = \mu \left[r \frac{d}{dr}\left(\frac{v_\theta}{r}\right) + \frac{1}{r}\underbrace{\frac{\partial v_r}{\partial \theta}}_{=0} \right] = \mu\, r \frac{d}{dr}\left(C_1 + \frac{C_2}{r^2}\right) = -\frac{2\mu C_2}{r^2} \tag{10}$$

The normal viscous stresses τ'_{rr} and $\tau'_{\theta\theta}$ are zero ($v_r = 0$, $\nabla \cdot \bar{v} = 0$), and the pressure force on a cylindrical surface is radial, so it has no moment about the axis. Only $\tau'_{r\theta}$ contributes to the torque.

Inner cylinder (surface $r = R_1$). The normal pointing into the liquid is $\bar{n} = +\bar{e}_r$, so $\bar{\bar{\tau}}' \cdot \bar{n} = \tau'_{r\theta}\, \bar{e}_\theta$ is the force per unit area that the liquid exerts on the cylinder. The torque per unit length about the axis, with $d\sigma = R_1\, d\theta\,(1)$ and arm $R_1 \bar{e}_r$:

$$\bar{M}_1 = \int_0^{2\pi} R_1 \bar{e}_r \wedge \tau'_{r\theta}(R_1) \bar{e}_\theta \, R_1 \, d\theta = 2\pi R_1^2 \, \tau'_{r\theta}(R_1)\, \bar{e}_z = -4\pi\mu C_2 \, \bar{e}_z \tag{11}$$

Outer cylinder (surface $r = R_2$). The normal pointing into the liquid is $\bar{n} = -\bar{e}_r$:

$$\bar{M}_2 = \int_0^{2\pi} R_2 \bar{e}_r \wedge [-\tau'_{r\theta}(R_2) \bar{e}_\theta] \, R_2 \, d\theta = -2\pi R_2^2 \, \tau'_{r\theta}(R_2)\, \bar{e}_z = +4\pi\mu C_2 \, \bar{e}_z \tag{12}$$

$$\boxed{\bar{M}_1 = -4\pi\mu \frac{(\Omega_1 - \Omega_2) R_1^2 R_2^2}{R_2^2 - R_1^2}\, \bar{e}_z = -\bar{M}_2} \tag{13}$$

The two torques are equal and opposite, as the angular momentum balance of the liquid requires: in steady motion the total moment on the liquid between the cylinders is zero, so the moments that the cylinders exert on it, $-\bar{M}_1$ and $-\bar{M}_2$, cancel.

3.5 Power needed to keep the cylinders rotating

The external agent must exert on each cylinder the opposite of the liquid torque, so the power it supplies is $\dot{W}_i = -\bar{M}_i \cdot \Omega_i \bar{e}_z$:

$$\dot{W}_1 = +4\pi\mu C_2\,\Omega_1, \qquad \dot{W}_2 = -4\pi\mu C_2\,\Omega_2 \tag{14}$$

$$\boxed{\dot{W} = \dot{W}_1 + \dot{W}_2 = 4\pi\mu C_2\,(\Omega_1 - \Omega_2) = 4\pi\mu \frac{(\Omega_1 - \Omega_2)^2 R_1^2 R_2^2}{R_2^2 - R_1^2}} \tag{15}$$

The total power is non-negative, as it must be: it equals the viscous dissipation in the gap (checked in 3.6).

3.6 Temperature distribution

With $v_r = 0$ the convective terms of (4.42) vanish. The energy equation for the steady, axisymmetric flow reads

$$0 = \phi_v + \frac{k}{r}\frac{d}{dr}\left(r\frac{dT}{dr}\right) \tag{16}$$

and the viscous dissipation of this flow (cylindrical ϕ_v of Notes.pdf, keeping the only non-zero shear $\tau'_{r\theta}$) is

$$\phi_v = \mu\left[r\frac{d}{dr}\left(\frac{v_\theta}{r}\right)\right]^2 = \mu\left(\frac{2C_2}{r^2}\right)^2 = \frac{4\mu C_2^2}{r^4} \tag{17}$$

Substituting (17) in (16) and multiplying by r/k:

$$\frac{d}{dr}\left(r\frac{dT}{dr}\right) = -\frac{4\mu C_2^2}{k\,r^3} \tag{18}$$

Integrating once, $r\,dT/dr = \dfrac{2\mu C_2^2}{k\,r^2} + B_1$; dividing by r and integrating again:

$$T(r) = -\frac{\mu C_2^2}{k\,r^2} + B_1 \ln r + B_2 \tag{19}$$

Write $a = \mu C_2^2/k$. The conditions $T(R_1) = T_1$ and $T(R_2) = T_2$ give, after eliminating B_2,

$$T - T_1 = a\left(\frac{1}{R_1^2} - \frac{1}{r^2}\right) + B \ln\frac{r}{R_1}, \qquad B \ln\frac{R_2}{R_1} = T_2 - T_1 - a\left(\frac{1}{R_1^2} - \frac{1}{R_2^2}\right) \tag{20}$$

$$\boxed{T(r) - T_1 = \frac{\mu C_2^2}{k}\left(\frac{1}{R_1^2} - \frac{1}{r^2}\right) + \left[T_2 - T_1 - \frac{\mu C_2^2}{k}\left(\frac{1}{R_1^2} - \frac{1}{R_2^2}\right)\right]\frac{\ln(r/R_1)}{\ln(R_2/R_1)}} \tag{21}$$

Check of the power. Integrating the dissipation over the gap per unit length, with $dV = 2\pi r\,dr$ and the antiderivative of r^{-3} equal to $-1/(2r^2)$:

$$\int \phi_v\,dV = 2\pi\int_{R_1}^{R_2} \frac{4\mu C_2^2}{r^4}\,r\,dr = 8\pi\mu C_2^2\left[-\frac{1}{2r^2}\right]_{R_1}^{R_2} = 4\pi\mu C_2^2\,\frac{R_2^2 - R_1^2}{R_1^2 R_2^2} = 4\pi\mu C_2\,(\Omega_1 - \Omega_2) \tag{22}$$

where $C_2(R_2^2 - R_1^2)/(R_1^2R_2^2) = \Omega_1 - \Omega_2$ from (6) was used. It coincides with (15): the power supplied by the cylinders is dissipated in the liquid and leaves as heat through the two walls.

PHASE 4

Interpretation, Limits and Dimensional Check

- **Physical meaning.** $v_\theta = C_1 r + C_2/r$ is a rigid rotation (the $C_1 r$ term) plus an irrotational vortex (the C_2/r term). Only the vortex part is sheared, with $\tau'_{r\theta} \propto 1/r^2$, so the stress is larger on the inner cylinder.

- **Limits.** For $\Omega_1 = \Omega_2 = \Omega$ the liquid rotates as a rigid body, $v_\theta = \Omega r$, $C_2 = 0$, the torques and the power vanish and $p - p(R_1) = \rho\Omega^2(r^2 - R_1^2)/2$. For $\Omega_2 = 0$ and $R_2 \to \infty$, $v_\theta = \Omega_1 R_1^2/r$ and $\bar{M}_1 = -4\pi\mu\Omega_1 R_1^2\,\bar{e}_z$, the viscous torque on a spinning cylinder in an unbounded liquid.
- **Dimensional check.** $v_\theta = C_1 r + C_2/r$: C_1 has units 1/s and C_2 has units $\mathrm{m^2/s}$. The torque per unit length $4\pi\mu C_2$ has units $(\mathrm{Pa\,s})(\mathrm{m^2/s}) = \mathrm{N}$, that is N m per metre of length. The power per unit length is N m/s per metre $=$ W/m. In (21), $\mu C_2^2/(k r^2)$ has units $(\mathrm{Pa\,s})(\mathrm{m^4/s^2})/[(\mathrm{W/(m\,K)})\,\mathrm{m^2}] = \mathrm{K}$.
- **Role of the dissipation.** If the viscous heating is negligible ($a \to 0$) the temperature is the purely logarithmic conduction profile between T_1 and T_2.

PROBLEM NS3

Spherical Balloon of Variable Radius

Original statement · NS3.pdf

Spherical symmetry | Unsteady flow | Inflation power | Energy balance

OFFICIAL PROBLEM STATEMENT

A spherical balloon of variable radius $R(t)$ is immersed in an infinite liquid medium of density ρ and viscosity μ. Far from the balloon, the liquid is at rest with pressure $p = p_\infty$. Determine the radial velocity $v_r(r,t)$ and pressure distribution $p(r,t)$ for the buoyancy-free spherically-symmetrical motion induced in the surrounding liquid. Obtain also the power required to inflate the balloon.

Additional question (handwritten on the sheet): where does the work go? Consider the following control volume.

The sheet carries no printed figure; the handwritten solution sketches the balloon of radius $r = R$ with surface velocity $v_r = \dot{R}$.

Source: NS3.pdf (page 1). Notes.pdf, Chapter 4: continuity Eq. 4.40, momentum Eq. 4.41 in spherical coordinates, mechanical-energy balance Eq. 4.24, viscous dissipation ϕ_v in spherical coordinates.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

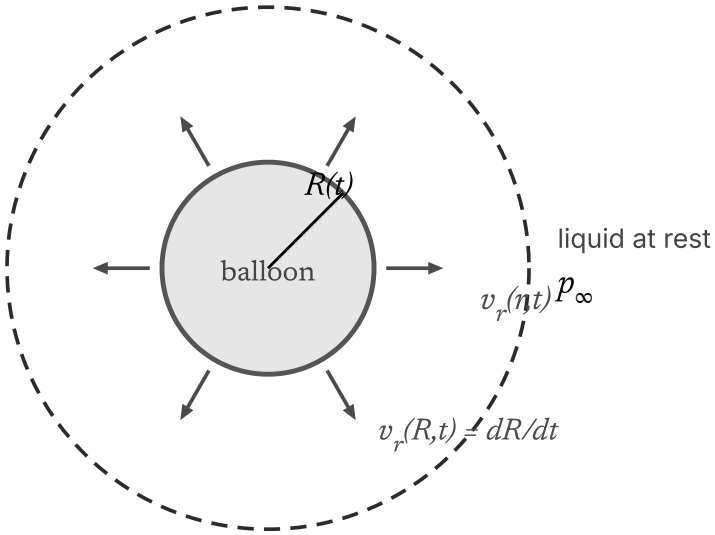
- Incompressible liquid with constant ρ and μ; equations (4.40) and (4.41) in spherical coordinates (r, θ, ϕ).
- Spherical symmetry: $\bar{v} = v_r(r,t)\,\bar{e}_r$, $v_\theta = v_\phi = 0$, and $\partial/\partial\theta = \partial/\partial\phi = 0$.
- Buoyancy-free: the mass force is neglected (or absorbed in the reduced pressure of Notes.pdf, Eq. 4.13).
- Unsteady: the balloon radius changes with time, so $\partial/\partial t \neq 0$.
- The liquid occupies $r > R(t)$; far from the balloon it is at rest with pressure p_∞.
- Data: $[\rho,\ \mu,\ p_\infty,\ R(t)]$. Unknowns: $v_r(r,t), p(r,t)$ and the inflation power $\dot{W}$.

Symbol	Meaning	SI unit
$R(t)$, $\dot{R}$, $\ddot{R}$	balloon radius, its rate of change and acceleration	m, m/s, m/s^2
ρ, μ	liquid density and viscosity	kg/m^3, Pa s
p_∞	pressure far from the balloon	Pa
$\dot{W}$	power delivered to the liquid by the balloon	W

PHASE 2

Frames, Field Form and Boundary Conditions

- Spherical frame (r, θ, ϕ) centred on the balloon centre, which stays fixed; $\bar{e}_r$ points away from the centre.
- Field form: $\bar{v} = v_r(r,t)\,\bar{e}_r, p = p(r,t)$.



control volume of the energy balance (dashed)

Balloon of radius $R(t)$ in the liquid: radial velocity $v_r(r,t) = R^2\dot{R}/r^2$, liquid at rest with pressure p_∞ far from the balloon, and the (dashed) control volume of the energy balance of Phase 3

Location	Condition	Origin
$r = R(t)$	$v_r = \dot{R}$	the liquid in contact with the balloon moves with its surface (kinematic condition)
$r \to \infty$	$v_r \to 0$	liquid at rest far from the balloon
$r \to \infty$	$p \to p_\infty$	statement

PHASE 3

Step-by-Step Derivation

3.1 Continuity and the radial velocity

$$\nabla \cdot \bar{v} = \frac{1}{r^2}\frac{\partial (r^2 v_r)}{\partial r} + \underbrace{\frac{1}{r\sin\theta}\frac{\partial (v_\theta \sin\theta)}{\partial \theta}}_{v_\theta = 0} + \underbrace{\frac{1}{r\sin\theta}\frac{\partial v_\phi}{\partial \phi}}_{v_\phi = 0} = 0 \implies r^2\, v_r = C(t) \tag{1}$$

The kinematic condition $v_r(R,t) = \dot{R}$ gives $C(t) = R^2\dot{R}$:

$$\boxed{v_r(r,t) = \frac{R^2\dot{R}}{r^2}} \tag{2}$$

The condition at infinity is satisfied automatically. The velocity at every point is fixed by the instantaneous radius and its rate of change.

3.2 Radial momentum and the pressure field

The viscous term of the r component of (4.41) is $\mu\,\partial/\partial r\left[r^{-2}\,\partial(r^2 v_r)/\partial r\right]$ (all other viscous terms contain v_θ, v_ϕ or derivatives with respect to θ, ϕ). By (1), $r^2 v_r = C(t)$ does not depend on r, so $\partial(r^2 v_r)/\partial r = 0$ and the viscous force vanishes identically: the flow is irrotational, $\nabla \wedge \bar{v} = 0$. The momentum equation reduces to

$$\rho\left(\frac{\partial v_r}{\partial t} + v_r\frac{\partial v_r}{\partial r}\right) = -\frac{\partial p}{\partial r} \tag{3}$$

The two accelerations are computed with the product and chain rules, using $d(R^2\dot{R})/dt = 2R\dot{R}^2 + R^2\ddot{R}$ and $\partial(r^{-2})/\partial r = -2r^{-3}$:

$$\frac{\partial v_r}{\partial t} = \frac{2R\dot{R}^2 + R^2\ddot{R}}{r^2}, \qquad v_r\frac{\partial v_r}{\partial r} = \frac{R^2\dot{R}}{r^2}\left(-\frac{2R^2\dot{R}}{r^3}\right) = -\frac{2R^4\dot{R}^2}{r^5} \tag{4}$$

$$\frac{\partial p}{\partial r} = -\rho\left[\frac{2R\dot{R}^2 + R^2\ddot{R}}{r^2} - \frac{2R^4\dot{R}^2}{r^5}\right] \tag{5}$$

Integrate (5) from r to ∞, with $p(\infty) = p_\infty$. The antiderivatives are $\int r'^{-2}dr' = -1/r'$ and $\int r'^{-5}dr' = -1/(4r'^4)$, and Barrow's rule between r and ∞ gives $\int_r^\infty r'^{-2}dr' = 1/r$ and $\int_r^\infty r'^{-5}dr' = 1/(4r^4)$:

$$p_\infty - p(r,t) = -\rho\left[\frac{2R\dot{R}^2 + R^2\ddot{R}}{r} - \frac{R^4\dot{R}^2}{2r^4}\right]$$

$$\boxed{p(r,t) = p_\infty + \rho\left[\frac{2R\dot{R}^2 + R^2\ddot{R}}{r} - \frac{R^4\dot{R}^2}{2r^4}\right]} \tag{6}$$

At the balloon surface $r = R$:

$$p(R,t) = p_\infty + \rho\left[2\dot{R}^2 + R\ddot{R} - \frac{\dot{R}^2}{2}\right] = p_\infty + \rho\left(\frac{3}{2}\dot{R}^2 + R\ddot{R}\right) \tag{7}$$

3.3 Viscous stress at the balloon surface

With the Navier-Poisson law and $\nabla \cdot \bar{v} = 0$, the radial normal stress is (Notes.pdf, stress components in spherical coordinates)

$$\tau'_{rr} = 2\mu\frac{\partial v_r}{\partial r} = 2\mu\left(-\frac{2R^2\dot{R}}{r^3}\right), \qquad \tau'_{rr}\Big|_{r=R} = -\frac{4\mu\dot{R}}{R} \tag{8}$$

The shear components $\tau'_{r\theta}$ and $\tau'_{r\phi}$ are zero by symmetry, so the viscous traction on the surface is purely radial.

3.4 Power required to inflate the balloon

The balloon surface moves with velocity $\dot{R}\,\bar{e}_r$. The liquid occupies $r > R$, so the unit normal pointing out of the liquid at the balloon surface is $-\bar{e}_r$. The traction that the balloon exerts on the liquid is $(-p\bar{\bar{I}} + \bar{\bar{\tau}}') \cdot (-\bar{e}_r) = (p - \tau'_{rr})\,\bar{e}_r$, and the power it delivers is the force times the velocity of the surface, integrated over the area $4\pi R^2$:

$$\dot{W} = 4\pi R^2\,\dot{R}\,[p(R,t) - \tau'_{rr}(R)] = 4\pi R^2 \dot{R}\,p(R,t) + 16\pi\mu R\dot{R}^2 \tag{9}$$

Substituting (7):

$$\boxed{\dot{W} = 4\pi R^2 \dot{R}\,p_\infty + 4\pi\rho R^2\dot{R}\left(\frac{3}{2}\dot{R}^2 + R\ddot{R}\right) + 16\pi\mu R\dot{R}^2} \tag{10}$$

3.5 Where does the work go? (additional question)

The three terms of (10) are identified by computing independently the rate of change of the kinetic energy of the liquid and its viscous dissipation.

Kinetic energy of the liquid. With $v_r^2 = R^4\dot{R}^2/r^4$ and $dV = 4\pi r^2\,dr$:

$$E_k = \int_R^\infty \frac{\rho}{2}\frac{R^4\dot{R}^2}{r^4}\,4\pi r^2\,dr = 2\pi\rho R^4\dot{R}^2\left[-\frac{1}{r}\right]_R^\infty = 2\pi\rho R^3\dot{R}^2 \tag{11}$$

Differentiating with the product rule,

$$\frac{dE_k}{dt} = 2\pi\rho\left(3R^2\dot{R}^3 + 2R^3\dot{R}\ddot{R}\right) = 4\pi\rho R^2\dot{R}\left(\frac{3}{2}\dot{R}^2 + R\ddot{R}\right) \tag{12}$$

which is exactly the second term of (10).

Viscous dissipation. In spherical coordinates with $v_\theta = v_\phi = 0$ the dissipation is $\phi_v = 2\mu\left[(\partial v_r/\partial r)^2 + 2(v_r/r)^2\right]$ (Notes.pdf, ϕ_v in spherical coordinates). With $\partial v_r/\partial r = -2R^2\dot{R}/r^3$ and $v_r/r = R^2\dot{R}/r^3$:

$$\phi_v = 2\mu\left[\frac{4R^4\dot{R}^2}{r^6} + \frac{2R^4\dot{R}^2}{r^6}\right] = \frac{12\mu R^4\dot{R}^2}{r^6} \tag{13}$$

$$\Phi = \int_R^\infty \frac{12\mu R^4\dot{R}^2}{r^6}\,4\pi r^2\,dr = 48\pi\mu R^4\dot{R}^2\left[-\frac{1}{3r^3}\right]_R^\infty = 16\pi\mu R\dot{R}^2 \tag{14}$$

which is the third term of (10).

Pressure term. $4\pi R^2\dot{R}\,p_\infty = p_\infty\,dV_b/dt$ with $V_b = 4\pi R^3/3$ the balloon volume: the work done against the ambient pressure to displace the liquid, that is, to set the ambient liquid in motion (the outward volume flux through any sphere is $4\pi r^2 v_r = 4\pi R^2\dot{R}$).

$$\boxed{\dot{W} = p_\infty\frac{dV_b}{dt} + \frac{dE_k}{dt} + \Phi} \tag{15}$$

The work supplied by the balloon increases the kinetic energy of the liquid, displaces the ambient pressure p_∞ and is partly converted into internal energy by viscous dissipation. This agrees with the mechanical energy balance (Notes.pdf, Eq. 4.24) applied to the liquid volume between the balloon and a distant material sphere.

PHASE 4

Interpretation, Limits and Dimensional Check

- **Physical meaning.** The velocity decays as $1/r^2$ because the volume flux $4\pi r^2 v_r$ through every sphere equals the rate of volume growth of the balloon. The motion is irrotational, so viscosity does not enter the pressure field (6); it appears only in the boundary traction (8) and the power (10).
- **Limits.** For $r \to \infty$, $v_r \to 0$ and $p \to p_\infty$. For a balloon at constant radius ($\dot{R} = 0$) every term vanishes: no flow, $p = p_\infty$ and no power. For a constant inflation rate ($\ddot{R} = 0$) the pressure is above p_∞ near the balloon, by the amount $\rho\dot{R}^2(3/2)$ at $r = R$.

- **Dimensional check.** v_r: $\mathrm{m^2}\cdot\mathrm{m/s/m^2} = \mathrm{m/s}$. In (6), $\rho R\dot{R}^2/r$ has units $(\mathrm{kg/m^3})(\mathrm{m^2/s^2}) = \mathrm{Pa}$ and $\rho R^2\ddot{R}/r$ has units $(\mathrm{kg/m^3})\,\mathrm{m^2}\,(\mathrm{m/s^2})/\mathrm{m} = \mathrm{Pa}$. In (10), $p_\infty\, 4\pi R^2\dot{R}$ is $\mathrm{Pa}\cdot\mathrm{m^3/s} = \mathrm{W}$ and $16\pi\mu R\dot{R}^2$ is $(\mathrm{Pa\,s})\,\mathrm{m}\,(\mathrm{m^2/s^2}) = \mathrm{W}$.
- **Consistency.** Equation (15) is verified term by term: (12) equals the second term of (10) and (14) equals the third, so the power computed from the surface traction equals the power accounted for inside the liquid.

PROBLEM NS4

Couette Flow with Heat Transfer

Original statement · NS4.pdf

Plane Couette flow | Viscous dissipation | Heat flux to the walls | Energy balance

OFFICIAL PROBLEM STATEMENT

Consider the steady planar motion of a perfect liquid of density ρ, viscosity μ, thermal conductivity k, and specific heat c in the gap of thickness h that forms between two infinite horizontal walls. The upper wall, at temperature T_u, moves parallel to itself with velocity U, while the lower wall remains at rest, at temperature T_l.

1. Determine the velocity components $v_x(z)$ and $v_z(z)$, as well as the accompanying pressure field $p(x, z)$, assuming that the pressure on the upper wall has the constant value $p = p_u$.

2. Calculate the volume flux for the flow established in the gap.

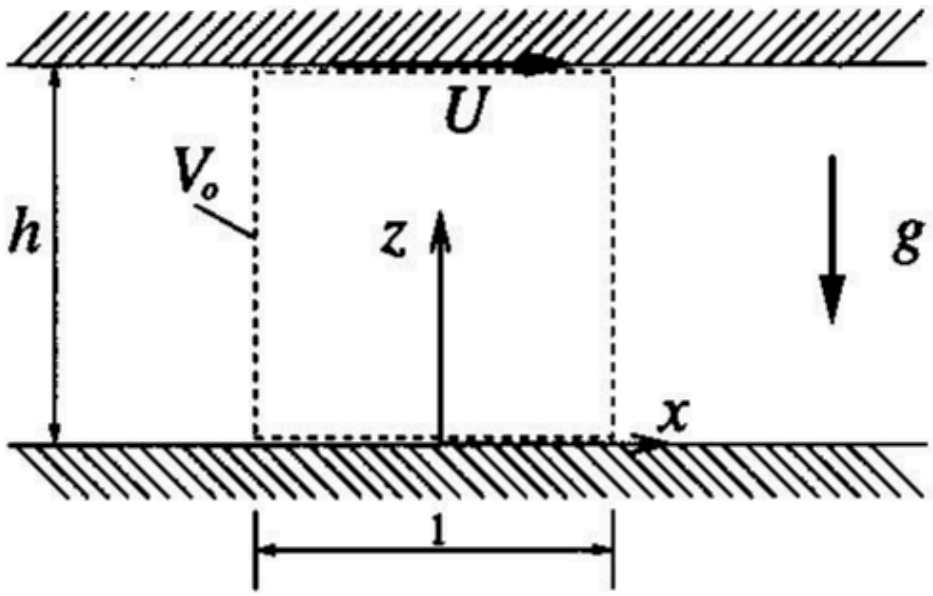
3. Obtain the pressure distribution $T(z)$.

4. Compute the force per unit surface acting on the upper wall as well as the power needed to move it.

5. Determine the conductive heat flux from the fluid to the upper and lower walls.

6. Use the fixed control volume V_o indicated in the figure to show that the power employed to move the upper wall is dissipated inside the liquid, generating an excess internal energy that is transfered by heat condution to the walls.

Transcribed as printed: in item 3 the sheet says "pressure distribution" where the temperature distribution $T(z)$ is meant, and "transfered" and "condution" are spelling errors of the sheet.



Statement figure (crop of NS4.pdf, p. 1): gap of thickness h, upper wall moving with U, fixed control volume V_o of unit length in x, axes x, z and gravity g

`Source: NS4.pdf (page 1). Notes.pdf, Chapter 4: continuity Eq. 4.40, momentum Eq. 4.41, energy Eq. 4.42, internal-energy balance Eq. 4.25, total-energy balance Eq. 4.3.`

Worked solution (4-phase method)

`PHASE 1`

Hypotheses, Data and Unknowns

- Steady planar flow of a perfect liquid (constant ρ, μ, k, c): equations (4.40)-(4.42).
- Infinite horizontal walls, gravity along $-\bar{e}_z$; no chemical reaction and no radiation ($Q_c = Q_r = 0$).
- The velocity and temperature depend only on the distance z to the lower wall: $\partial/\partial x = \partial/\partial y = 0$ for them. The pressure may depend on x and z until it is determined.
- The walls are impermeable. The lower wall ($z = 0$) is at rest at T_l; the upper wall ($z = h$) moves with $U\,\bar{e}_x$ at T_u and the pressure on it is p_u.
- Data: $[\rho,\ \mu,\ k,\ c,\ h,\ U,\ T_l,\ T_u,\ p_u,\ g]$. Unknowns: $v_x(z), v_z(z), p(x,z), T(z)$, the flow rate, the wall force and power, the wall heat fluxes.

- Degrees of freedom: two velocity components, pressure and temperature, each a function of z (the pressure also of x).

Symbol	Meaning	SI unit
h	gap between the walls	m
U	velocity of the upper wall	m/s
T_l, T_u	temperatures of the lower and upper walls	K
p_u	pressure on the upper wall	Pa
ρ, μ, k, c	density, viscosity, conductivity, specific heat	kg/m^3, Pa s, W/(m K), J/(kg K)

PHASE 2

Frames, Field Form and Boundary Conditions

- Cartesian frame (x, y, z): x along the walls in the direction of U, z normal to the walls and upward from the lower wall, y out of the plane. The basis is orthonormal and right-handed (no change of basis).
- Field form: $\bar{v} = v_x(z)\,\bar{e}_x + v_z(z)\,\bar{e}_z, T = T(z), p = p(x, z)$.
- Fixed control volume V_o of the figure: the box $0 < z < h$ of unit length in x and unit width in y; its surfaces are the lower wall Σ_l ($\bar{n} = -\bar{e}_z$), the upper wall Σ_u ($\bar{n} = +\bar{e}_z$) and the two vertical faces Σ_1 ($\bar{n} = -\bar{e}_x$) and Σ_2 ($\bar{n} = +\bar{e}_x$).

Location	Condition	Origin
$z = 0$	$v_x = 0, v_z = 0, T = T_l$	lower wall at rest, impermeable
$z = h$	$v_x = U, v_z = 0, T = T_u$	upper wall moving, impermeable
$z = h$	$p = p_u$	statement

PHASE 3

Step-by-Step Derivation

3.1 Continuity and the normal velocity

$$\frac{\partial v_x}{\partial x} + \frac{dv_z}{dz} = \underbrace{\frac{\partial v_x}{\partial x}}_{v_x = v_x(z)} + \frac{dv_z}{dz} = 0 \implies v_z = \text{const} \qquad (1)$$

The wall is impermeable, $v_z(0) = 0$, so

$$\boxed{v_z(z) = 0} \tag{2}$$

3.2 Momentum equations

z-component: with $v_z = 0$ every inertial and viscous term vanishes,

$$0 = -\frac{\partial p}{\partial z} - \rho g \implies p + \rho g z = f(x) \tag{3}$$

x-component: the convective terms vanish ($\partial_x = 0$, $v_z = 0$),

$$\rho\Big(\underbrace{\cancel{\frac{\partial v_x}{\partial t}}}_{\text{steady}} + \underbrace{v_x\cancel{\frac{\partial v_x}{\partial x}}}_{\partial_x = 0} + \underbrace{v_z\cancel{\frac{\partial v_x}{\partial z}}}_{v_z = 0}\Big) = -\frac{\partial p}{\partial x} + \mu\frac{d^2 v_x}{dz^2} \implies \mu\frac{d^2 v_x}{dz^2} = \frac{\partial p}{\partial x} = f'(x) \tag{4}$$

The left side depends only on z and the right side only on x, so both are equal to a constant. The pressure on the upper wall is $p(x, h) = p_u$ for every x, and by (3) $f(x) = p_u + \rho g h$ is a constant, so $f'(x) = 0$:

$$\boxed{p(x, z) = p_u + \rho g\,(h - z)}, \qquad \frac{\partial p}{\partial x} = 0 \tag{5}$$

Equation (4) becomes $d^2 v_x / dz^2 = 0$; integrating twice, $v_x = C_1 + C_2 z$. The conditions $v_x(0) = 0$ and $v_x(h) = U$ give $C_1 = 0$ and $C_2 = U/h$:

$$\boxed{v_x(z) = U\,\frac{z}{h}} \tag{6}$$

3.3 Volume flux

The flux per unit width in y through a section $x = \text{const}$ is $Q = \int_0^h v_x \, dz$. The antiderivative of Uz/h is $Uz^2/(2h)$ and Barrow's rule gives:

$$\boxed{Q = \left[\frac{U}{h} \frac{z^2}{2} \right]_0^h = \frac{Uh}{2}} \tag{7}$$

3.4 Temperature distribution

The energy equation (4.42) with $v_z = 0$, $\partial_x = 0$ and $\phi_v = \mu (dv_x/dz)^2 = \mu U^2/h^2$:

$$\underbrace{\rho c v_x \frac{\partial T}{\partial x}}_{\partial_x = 0} = \phi_v + k \frac{d^2 T}{dz^2} \quad \Longrightarrow \quad \frac{d^2 T}{dz^2} = -\frac{\mu U^2}{k \, h^2} \tag{8}$$

Integrating twice, $T = B_1 + B_2 z - \dfrac{\mu U^2}{2 k h^2} z^2$. The conditions $T(0) = T_l$ and $T(h) = T_u$ give $B_1 = T_l$ and $B_2 = \dfrac{T_u - T_l}{h} + \dfrac{\mu U^2}{2kh}$:

$$\boxed{T(z) = T_l + (T_u - T_l) \, \frac{z}{h} + \frac{\mu U^2}{2k} \, \frac{z}{h} \left(1 - \frac{z}{h} \right)} \tag{9}$$

3.5 Force on the upper wall and power needed to move it

On the upper wall, the normal pointing into the liquid is $\bar{n} = -\bar{e}_z$, so $\left(-p \bar{\bar{I}} + \bar{\bar{\tau}}' \right) \cdot \bar{n}$ is the force per unit area that the liquid exerts on the wall. The viscous components are $\tau'_{xz} = \mu \, dv_x/dz = \mu U/h$ and $\tau'_{zz} = 0$ (Notes.pdf, stress components, Cartesian coordinates):

$$\bar{f}_u = -p_u (-\bar{e}_z) + \bar{\bar{\tau}}' \cdot (-\bar{e}_z) = p_u \, \bar{e}_z - \mu \frac{U}{h} \, \bar{e}_x \tag{10}$$

The viscous part opposes the motion of the wall. The external agent must supply $+\mu U/h \, \bar{e}_x$ per unit area, with the wall velocity $U \bar{e}_x$:

$$\boxed{\bar{f}_u = p_u\,\bar{e}_z - \mu\frac{U}{h}\,\bar{e}_x, \qquad \dot{W} = \mu\frac{U^2}{h}\ \text{per unit area}} \tag{11}$$

The pressure force $p_u\bar{e}_z$ is normal to the wall and does no work on it.

3.6 Heat fluxes to the walls

The heat flux from the liquid to a wall is $\bar{q}\cdot\bar{n}$, with $\bar{n}$ pointing out of the liquid and $\bar{q} = -k\nabla T$. From (9),

$$\frac{dT}{dz} = \frac{T_u - T_l}{h} + \frac{\mu U^2}{2kh} - \frac{\mu U^2}{kh^2}z \tag{12}$$

Upper wall ($\bar{n} = +\bar{e}_z$, $z = h$):

$$q_u = -k\frac{dT}{dz}\Big|_{z=h} = k\frac{T_l - T_u}{h} + \frac{\mu U^2}{2h} \tag{13}$$

Lower wall ($\bar{n} = -\bar{e}_z$, $z = 0$):

$$q_l = +k\frac{dT}{dz}\Big|_{z=0} = k\frac{T_u - T_l}{h} + \frac{\mu U^2}{2h} \tag{14}$$

$$\boxed{q_u = k\frac{T_l - T_u}{h} + \frac{\mu U^2}{2h}, \qquad q_l = k\frac{T_u - T_l}{h} + \frac{\mu U^2}{2h}} \tag{15}$$

The conductive contributions are equal and opposite and the viscous heating splits equally between the walls. Their sum is $q_u + q_l = \mu U^2/h$.

3.7 Integral balance in the fixed control volume V_o

Total energy (Notes.pdf, Eq. 4.3 in integral form for the fixed volume, steady):

$$\int_{\Sigma_o} \rho\left(e+\frac{|\bar v|^2}{2}\right)\bar v\cdot\bar n\,d\sigma = -\int_{\Sigma_o} p\,\bar v\cdot\bar n\,d\sigma + \int_{\Sigma_o}\bar v\cdot\bar{\bar\tau}'\cdot\bar n\,d\sigma + \underbrace{\int_{V_o}\rho\bar g\cdot\bar v\,dV}_{=0} - \int_{\Sigma_o}\bar q\cdot\bar n\,d \tag{16}$$

Each term, surface by surface (areas of unit length times unit width):

Surface	**Convective** $\int \rho(e+v^2/2)\,\bar v\cdot\bar n$	**Pressure work** $-\int p\,\bar v\cdot\bar n$	**Viscous work** $\int \bar v\cdot\bar{\bar\tau}'\cdot\bar n$	**Heat** $-\int \bar q\cdot\bar n$
Σ_l ($z=0$)	0 ($\bar v\cdot\bar n=0$)	0	0 ($\bar v=0$)	$-q_l$
Σ_u ($z=h$)	0 ($\bar v\cdot\bar n=0$)	0	$U\,\tau'_{xz}=\mu U^2/h$	$-q_u$
$\Sigma_1+\Sigma_2$	the two faces cancel ($\bar v$, e, p do not depend on x; opposite normals)	cancel	cancel	0 (T independent of x)

The mass-force work vanishes because $\bar g\cdot\bar v=-g\,v_z=0$. Equation (16) gives

$$0=\mu\frac{U^2}{h}-q_u-q_l \;\Longrightarrow\; \boxed{\mu\frac{U^2}{h}=q_u+q_l} \tag{17}$$

Internal energy (Notes.pdf, Eq. 4.25, steady, $\nabla\cdot\bar v=0$): $0=-\int_{\Sigma_o}\bar q\cdot\bar n\,d\sigma+\int_{V_o}\phi_v\,dV$, with $\int_{V_o}\phi_v\,dV=\dfrac{\mu U^2}{h^2}\,h=\dfrac{\mu U^2}{h}$ (area 1, height h), which again gives $q_u+q_l=\mu U^2/h$.

Equations (11), (15) and (17) show the whole chain: the power spent moving the upper wall, $\mu U^2/h$ per unit area, equals the viscous dissipation in the liquid and leaves the volume as the heat conducted to the two walls.

PHASE 4

Interpretation, Limits and Dimensional Check

- **Physical meaning.** The pressure is hydrostatic and the velocity is linear: viscous forces alone transmit the motion. The dissipation $\phi_v=\mu U^2/h^2$ is uniform, and it heats the liquid; the parabolic term of (9) is the temperature rise it produces, with its maximum in

the middle of the gap.

- **Limits.** For $U \to 0$ there is no flow and no dissipation: T is the linear conduction profile between T_l and T_u and $q_u = -q_l = k(T_l - T_u)/h$. For $T_u = T_l$, the temperature is symmetric about $z = h/2$ and each wall receives $\mu U^2/(2h)$.
- **Dimensional check.** $\mu U/h$: $(\mathrm{Pa\,s})(\mathrm{m/s})/\mathrm{m} = \mathrm{Pa}$. $\mu U^2/h$: $\mathrm{Pa \cdot m/s} = \mathrm{W/m^2}$. In (9), $\mu U^2/(2k)$ has units $(\mathrm{Pa\,s})(\mathrm{m^2/s^2})/[\mathrm{W/(m\,K)}] = \mathrm{K}$.
- **Mean velocity.** From (7), $\bar{v} = Q/h = U/2$, half the wall velocity.